FINAL RESEARCH PROJECT - PROCESS REPORT

HPC Jobs, Resources, and Capacity Audit

A point-in-time operational record of submissions, resource use, storage headroom, and the decision not to launch full screening.

5	1	0.309 GiB
V2 JOB SUBMISSIONS TO DATE	NEW JOB IN CUTOVER WINDOW	POST-SMOKE SOFT MARGIN

OPERATOR DECISION

Five V2 jobs exist through job 3550; only one was submitted in the storage-safe cutover window. The queue was empty after smoke completion. The post-smoke capacity pass had only about 0.309 GiB margin and has expired. Full screening remains unsubmitted.

TECHNICAL SUMMARY

The operational bottleneck is group-level storage, not compute credit or the project-ID quota. The storage-safe policy reduced the projected requirement enough to pass the 17 July point-in-time gate, but with a narrow margin. Job 3550 completed successfully and the project deliberately stopped. This report is an audit record, not a standing authorization to submit the full screening array.

Campaign	large_scale_thai_drl_v2
Evidence cut	17 July 2026
Generated	20 July 2026
Audience	Research team, technical reviewer, and HPC operator

Evidence basis: Slurm accounting for jobs 3540, 3541, 3543, and 3550; the quarantined job 3542 incident receipt; and signed pre- and post-smoke capacity decisions.

SECTION 1

Submission register and resource evidence

Job	Purpose	State	Elapsed	CPUs	Memory	Evidence role
3540	Vector calibration	COMPLETED	07:45	18	64G	Accepted calibration
3541	Determinism	COMPLETED	00:07	2	4G	Accepted reproducibility
3542	Original screening smoke	FAILED	00:20	2	8G	Quarantined incident
3543	Replacement screening smoke	COMPLETED	33:44	2	8G	Accepted baseline smoke
3550	Storage-safe screening smoke	COMPLETED	33:20	2	8G	Accepted cutover verification

SUBMISSION-COUNT INTERPRETATION

Jobs 3540 through 3543 were four submissions in the earlier V2 pass. Job 3550 was the sole submission in the later storage-safe cutover window. The full screening array was not submitted.

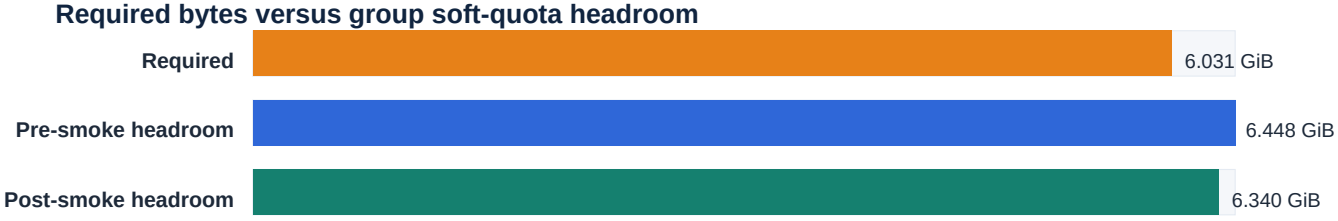
JOB 3550 RESOURCE DETAIL

- Scheduler outcome: COMPLETED, exit 0:0, elapsed 00:33:20.
- Allocation: 2 CPUs and 8 GiB requested memory on compute-devel.
- Observed batch MaxRSS: 809,680 KiB; TotalCPU: 32:33.567.
- The job produced five complete manifests, five retention receipts, zero checkpoint paths, and a passing strict import.

SECTION 2

Capacity decision before and after the smoke run

Required bytes versus group soft-quota headroom



The same 6.031 GiB requirement includes a 1.5 safety factor plus a fixed 0.250 GiB reserve.

Capacity item	Pre-smoke	Post-smoke	Interpretation
Decision	PASS	PASS	All eight byte/inode checks passed
Group soft headroom	6.448 GiB	6.340 GiB	Headroom decreased after outputs arrived
Required bytes	6.031 GiB	6.031 GiB	Safety-adjusted retained projection plus reserve
Margin	427.340 MiB	316.101 MiB	Only about 0.417 GiB before and 0.309 GiB after
Write probe	Passed and removed	Passed and removed	1 MiB fsync test did not leave a probe file
Validity	15 minutes	15 minutes	Both point-in-time decisions are now expired

CHECK MATRIX

Check	Pre	Post
group bytes fit hard headroom	PASS	PASS
group bytes fit soft headroom	PASS	PASS
group inodes fit hard headroom	PASS	PASS
group inodes fit soft headroom	PASS	PASS
project bytes fit hard headroom	PASS	PASS
project bytes fit soft headroom	PASS	PASS
project inodes fit hard headroom	PASS	PASS
project inodes fit soft headroom	PASS	PASS

SECTION 3

Credits, projection, and operating guardrails

Resource	Point-in-time value	Assessment
Project compute credit	9,979.88 SHr remaining (99.80%)	Healthy; not the immediate blocker
Personal compute credit	496.25 SHr remaining (99.25%)	Healthy for bounded work
Project-ID storage	About 36.5 GiB used of 300 GiB hard limit	Healthy; does not override inherited group quota
Personal-group storage	6.340 GiB soft headroom after smoke	Narrow margin; durable ownership/quota fix preferred

Projection component	Bytes	GiB
Original checkpoint-writing full screen	8,255,696,460	7.6887
Storage-safe retained outputs	4,138,125,200	3.8539
Checkpoint bytes avoided	4,123,243,380	3.8401
Required after safety factor and reserve	6,475,623,256	6.0309

NEXT-WINDOW GUARDRAILS

- Capture a new signed quota snapshot immediately before rendering any full-screening submission.
- Require all eight byte and inode checks, the write probe, release identity, preflight identity, and runtime projection checks to pass.
- Use no more than four new sbatch submissions in the window; the recommended screening sequence needs one array submission and, after verification, one aggregate submission.
- Stop on any quota drift, mismatched release, nonempty conflicting queue, failed task, missing receipt, or unexpected checkpoint path.
- Prefer correcting SGID/group ownership or raising the personal-group quota before the full run; the 0.309 GiB margin is operationally fragile.

STANDING DECISION

No full-screening authorization survives from the 17 July snapshot. The operator must reopen the gate with fresh evidence.

AUDIT SOURCES

Artifact or identifier	Verified value
Pre-smoke capacity SHA-256	`3a53801913a7529f 26263b4be57d7011 ed59b1517fa17d86 01e40ea1a1d0059d`
Post-smoke capacity SHA-256	`52b7ae257896b89e 1a372805533939c2 b68239b8bc05224d 6882f26f2ede1258`
Post-smoke decision SHA-256	`ffbcb457cb055c5d 07bd0ed92447fdab 1d2400672d91b114 46bb30df47929121`
Job 3550 accounting SHA-256	`6959a20d6a95d7f0 a8d57852f40302d6 6233f587d62fe9c3 29848cf2a4608601`